

系统“希望特性”设计方法

常规设计: $L_0(\omega) + L_c(\omega) = L(\omega)$

希望特性设计: $L(\omega) - L_0(\omega) = L_c(\omega)$

例 2 已知单位反馈系统的开环传递函数为:

$$G_0(s) = \frac{K}{s(\frac{s}{10} + 1)(\frac{s}{50} + 1)(\frac{s}{100} + 1)(\frac{s}{200} + 1)}$$

$$\text{要求: } \begin{cases} r(t) = t \rightarrow e_{ss}^* \leq 1/200 \\ \sigma \% \leq 30 \% \\ t_s \leq 0.7 \end{cases}$$

试确定 $G_c(s)$

$$\text{解. (1) 由要求: } \begin{cases} r(t) = t \rightarrow e_{ss}^* \leq 1/200 \\ \sigma \% \leq 30 \% \\ t_s \leq 0.7 \end{cases}$$

$$\text{查 P173 图 5-56} \Rightarrow \begin{cases} K = 200 \\ \gamma^* \geq 47^\circ \\ t_s = 9/\omega_c^* \leq 0.7 \rightarrow \omega_c^* \geq 9/0.7 = 13 \end{cases}$$

$$(2) \text{ 绘制 } L_0(\omega) \rightarrow \omega_{c0} = \sqrt{10 \times 200} = 44.72$$

(3) “希望特性”设计

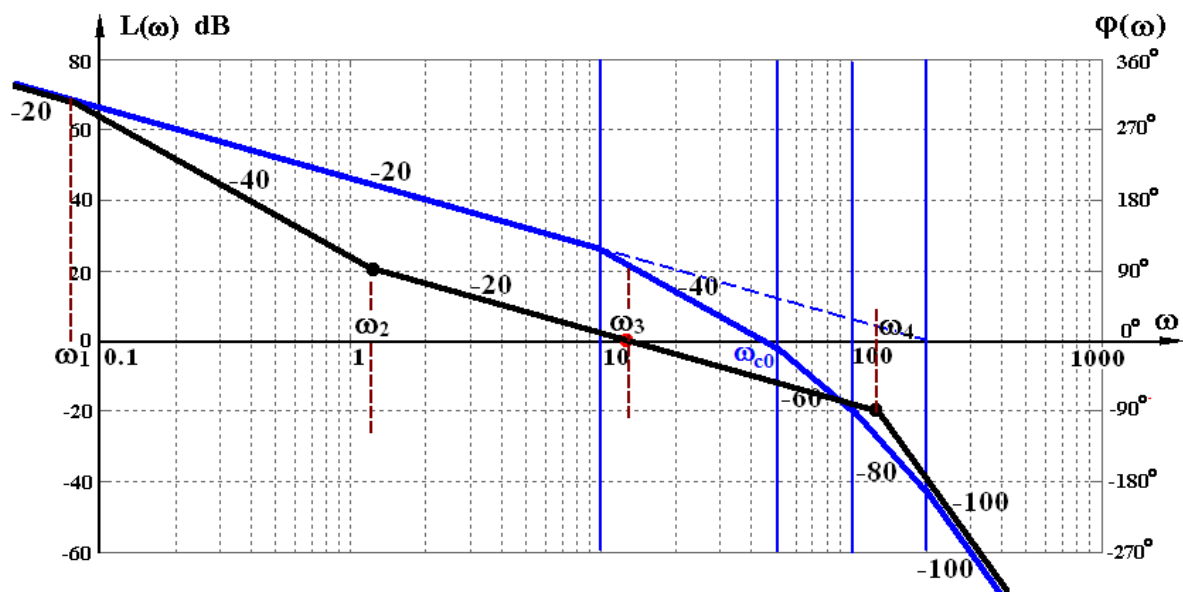
① 在 $\omega_c^* = 13$ 做 -20dB/dec 直线, 向两端各延伸十倍频;

② 在 $\omega_2 = 0.1\omega_c^* = 1.3$ 处做 -40dB/dec 直线与 $L_0(\omega)$ 低频段相接于 $\omega_1 = 0.0845$;

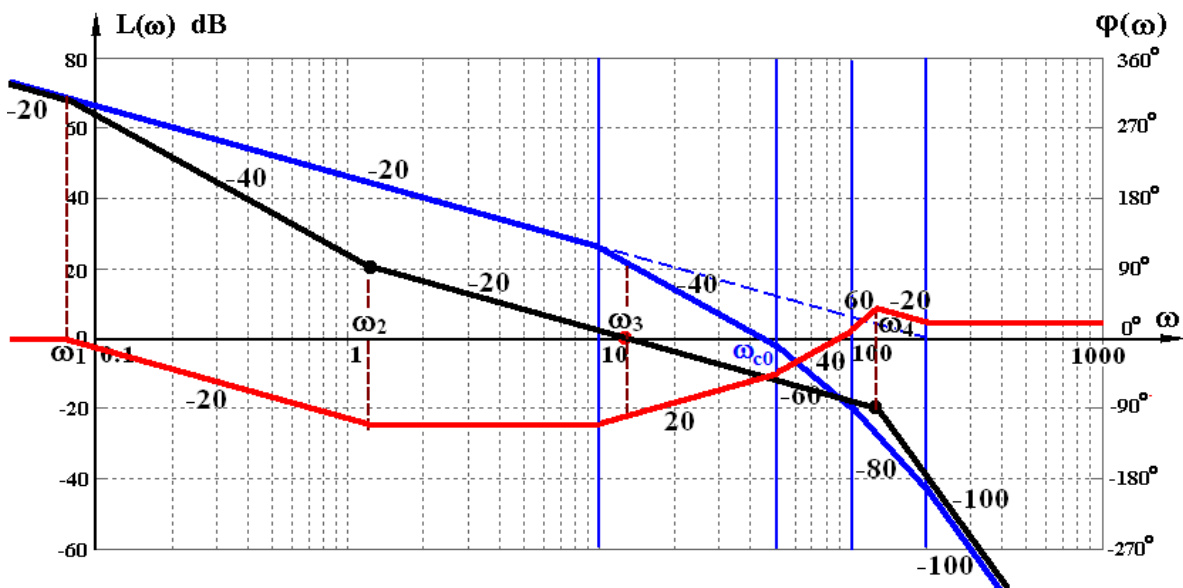
③ 在 $\omega_4 = 130$ 处做 -100dB/dec 直线 (与 $L_0(\omega)$ 高频段同斜率);

$$G_0(s) = \frac{K}{s(\frac{s}{10} + 1)(\frac{s}{50} + 1)(\frac{s}{100} + 1)(\frac{s}{200} + 1)}$$

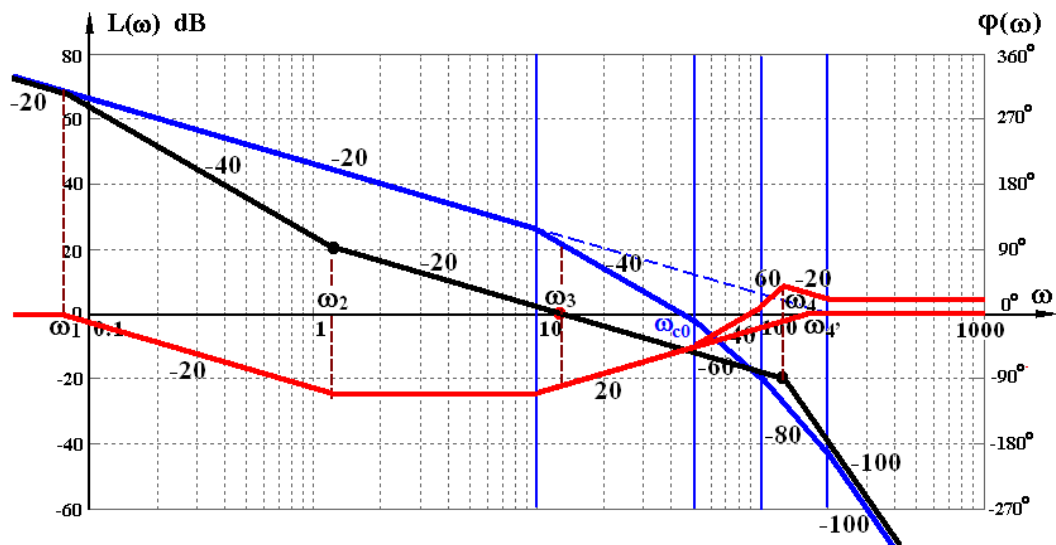
$$\begin{cases} K = 200 \\ \gamma^* \geq 47^\circ \\ \omega_c^* \geq 13 \end{cases}$$



$$(4) L_{c1}(\omega) = L_1(\omega) - L_0(\omega)$$



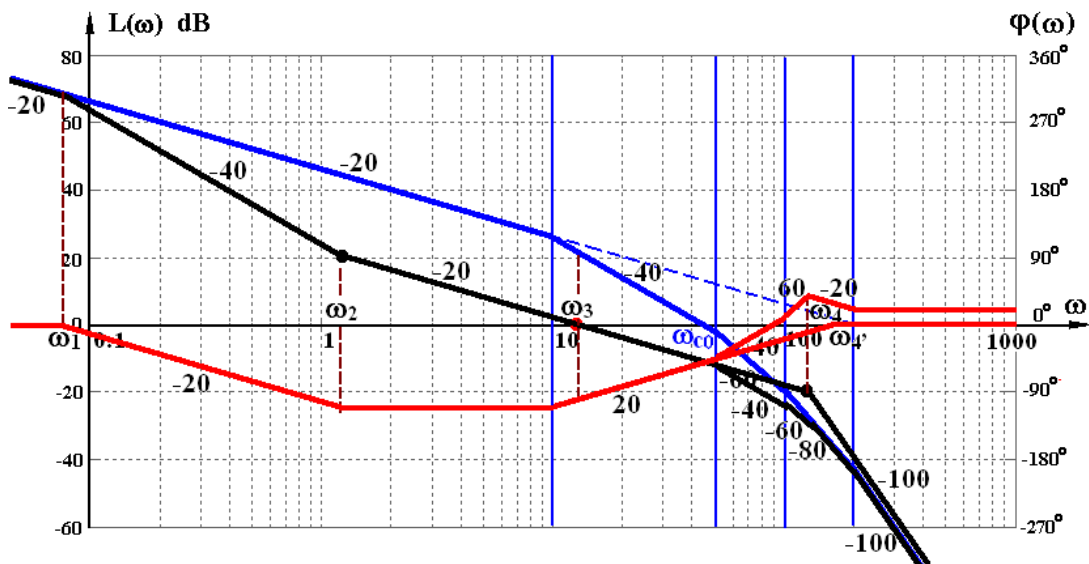
中-高频段交接处太复杂 → 简化



$$\frac{\omega_4}{\omega_{c0}} = \frac{\omega_{c0}}{\omega_c}, \quad \omega_4 = \frac{\omega_{c0}^2}{\omega_c} = \frac{44.72^2}{13} = 154$$

$$\frac{\omega_4}{10} = \frac{\omega_2}{\omega_1}, \quad \omega_1 = \frac{10\omega_2}{\omega_4} = \frac{13}{154} = 0.0844$$

得出 $G_c(s) = \frac{(\frac{s}{1.3} + 1)(\frac{s}{10} + 1)}{(\frac{s}{0.0844} + 1)(\frac{s}{154} + 1)}$ (如图所示)



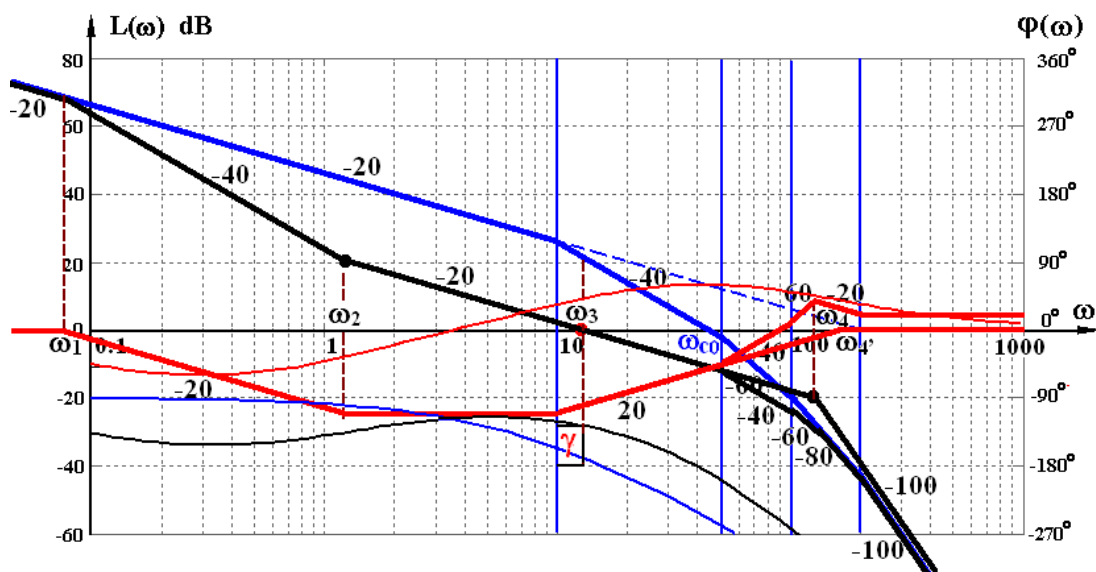
$$G(s) = \frac{(\frac{s}{1.3} + 1)(\frac{s}{10} + 1)}{(\frac{s}{0.0844} + 1)(\frac{s}{154} + 1)} \cdot \frac{200}{s(\frac{s}{10} + 1)(\frac{s}{50} + 1)(\frac{s}{100} + 1)(\frac{s}{200} + 1)}$$

$$= \frac{200(\frac{s}{1.3} + 1)}{s(\frac{s}{50} + 1)(\frac{s}{100} + 1)(\frac{s}{200} + 1)(\frac{s}{0.0844} + 1)(\frac{s}{154} + 1)}$$

(5) 验算:

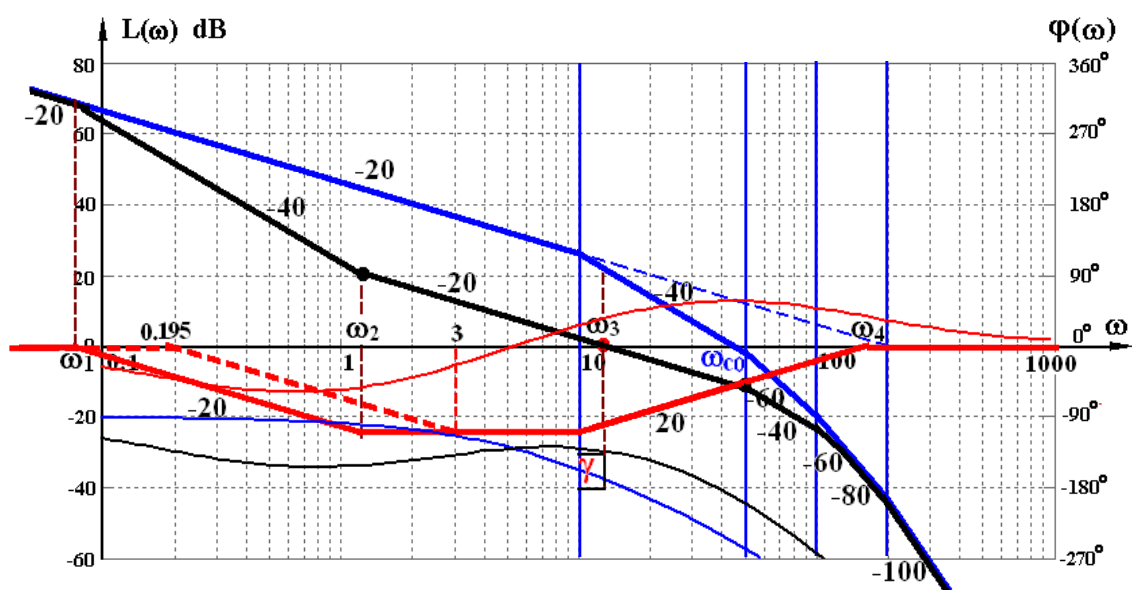
$$\omega_c = 13$$

$$\begin{aligned} \gamma &= 180^\circ + \angle \varphi(\omega_c) \\ &= 180^\circ + \arctan \frac{13}{1.3} - 90^\circ - \arctan \frac{13}{50} - \arctan \frac{13}{100} \\ &\quad - \arctan \frac{13}{200} - \arctan \frac{13}{0.0844} - \arctan \frac{13}{154} \\ &= 180^\circ + 82.29^\circ - 90^\circ - 14.57^\circ - 7.41^\circ - 3.72^\circ - 89.63^\circ - 4.83^\circ \\ &= 54.1^\circ > 47^\circ = \gamma^* \end{aligned}$$



(6) 考虑具体情况进行调整：

① 为便于工程实现，将迟后段右移（虚线所示）



$$\omega_2 = 1.3 \rightarrow 3$$

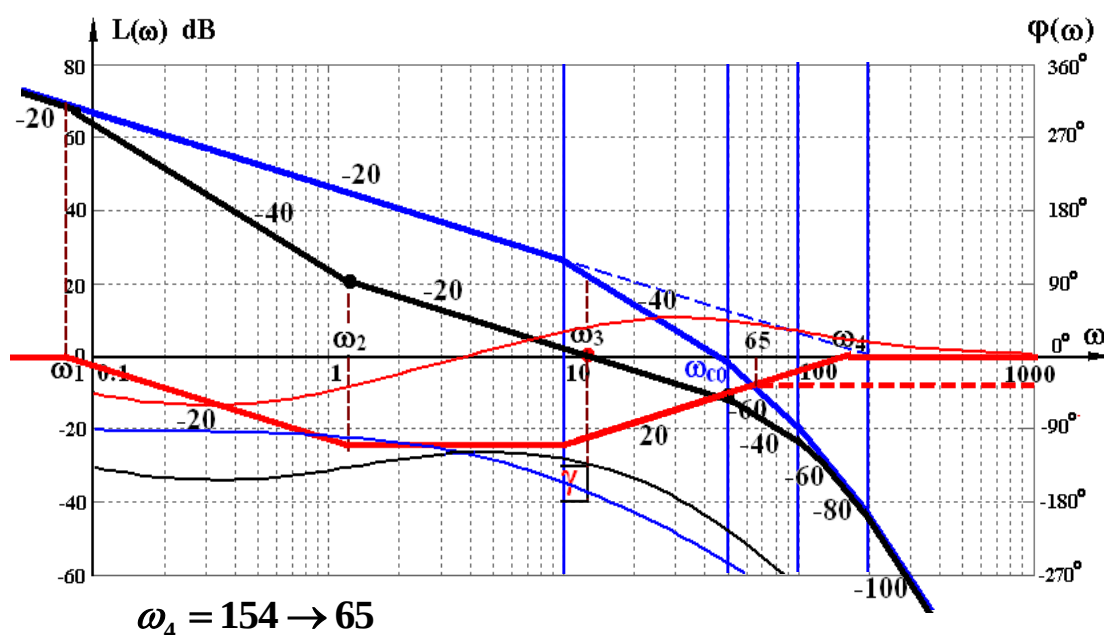
$$\omega_1 = 0.0844 \rightarrow 0.195$$

$$G_{c1}(s) = \frac{\left(\frac{s}{3} + 1\right)\left(\frac{s}{10} + 1\right)}{\left(\frac{s}{0.195} + 1\right)\left(\frac{s}{154} + 1\right)}$$

$$\omega_c = 13$$

$$\begin{aligned} \gamma &= 180^\circ + \angle \varphi(\omega_c) \\ &= 180^\circ + \arctan \frac{13}{3} - 90^\circ - \arctan \frac{13}{50} - \arctan \frac{13}{100} \\ &\quad - \arctan \frac{13}{200} - \arctan \frac{13}{0.195} - \arctan \frac{13}{154} \\ &= 180^\circ + 77^\circ - 90^\circ - 14.57^\circ - 7.41^\circ - 3.72^\circ - 89.14^\circ - 4.83^\circ \\ &= 47.33^\circ > 47^\circ = \gamma^* \end{aligned}$$

为提高系统抗高频干扰能力，可将高频段压低



$$G_{c2}(s) = \frac{\left(\frac{s}{1.3} + 1\right)\left(\frac{s}{10} + 1\right)}{\left(\frac{s}{0.0845} + 1\right)\left(\frac{s}{65} + 1\right)}$$

$$\omega_c = 13$$

$$\begin{aligned} \gamma &= 180^\circ + \angle \varphi(\omega_c) \\ &= 180^\circ + \arctan \frac{13}{1.3} - 90^\circ - \arctan \frac{13}{50} - \arctan \frac{13}{100} \\ &\quad - \arctan \frac{13}{200} - \arctan \frac{13}{0.0845} - \arctan \frac{13}{65} \\ &= 180^\circ + 84.29^\circ - 90^\circ - 14.57^\circ - 7.41^\circ - 3.72^\circ - 89.63^\circ - 11.31^\circ \\ &= 47.65^\circ > 47^\circ = \gamma^* \end{aligned}$$